

Figure 1. Frozen architecture overview

	Dense	BM25	Fusion	Reranker	Final context	LLM
P0	-	-	-	-	-	gpt-4o-mini
P1	top 5	-	-	-	5	gpt-4o-mini
P2	top 10	top 10	RRF k=60 top 5	-	5	gpt-4o-mini
P3	top 20	top 20	RRF k=60 top 20	Cross-encoder 512 tokens	5	gpt-4o-mini

Frozen run: financebench-heldout-v1 | Shared output limit: 256 tokens | P1-P3 use one RAG prompt builder